Looking Away

Building 2: Loops

Ways between ways, and the systems that come back.

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A later chapter builds a system and blurs it by the action of an earlier one; a *loop* is the theorem saying the earlier system comes back (rule F8). Building 2 also holds the levels above moves: ways between ways (Chapter 16 is its first room).

Chapter numbers restart in this book; Building 0 and Building 1 are cited by name.

Part I is built and blurred without imports. What used to be Part II, the edge (loops on the map, Amplify, atoms, the missing move), is research rather than floors; it is kept as a separate document, *Field notes: the edge* (`fieldnotes-edge.pdf`).

Import ledger (rule C8). Imported lines: *none*. Every Given line cites an earlier chapter; Chapter 2 builds the rules of $\Box$ itself (D2) instead of importing them from arithmetic. Reported facts in Remarks, used by no line: Chapter 1, R1 (pigeonhole proof length), R2 (completeness with "for all"), R3 (incompleteness); Chapter 2, R2 (the arrow into arithmetic), R3 (ε_0 ladder).

Part I. Words about words

Building 0, Chapter 0 writes rules about words; Building 1, Chapter 2 builds words and glues them. Here the words are statements and proofs, and the blurs are "forget the proof" and "forget everything but the answers". Both chapters are built and blurred inside the part: nothing is imported. Arithmetic, where the famous theorems live, is reached only by arrows in the Remarks, and those arrows are marked reported.

Chapter 1. Proofs

statements → what every world says · from: Building 0, Chapters 0, 3; Building 1, Chapter 2 · build, blur

Statements and proofs are words. Glue two statements when each proves the other; glue them when every world gives them the same answer. The two blurs agree.

Given

G1 (Building 0, Chapter 0, M3.) Words in a list of names; maps on words by recursion.

G2 (Building 1, Chapter 2.) Words, glued by rules from below.

G3 (Building 0, Chapter 3.) $\{0, 1\}$ as $\mathbb{Z}/2$.

Definitions

D1 (Build, M3.) *Statements*: words in letters $p_1, \ldots, p_n$, the name $\bot$, and $\neg$, $\wedge$, $\vee$, $\to$. Write $A \leftrightarrow B$ for $(A \to B) \wedge (B \to A)$.

D2 (Build.) A *proof* of A: a list of statements ending in A, each one an instance of $A \to (B \to A)$, $(A \to (B \to C)) \to ((A \to B) \to (A \to C))$, $(\neg B \to \neg A) \to (A \to B)$ (with $\neg A := A \to \bot$ and $\wedge$, $\vee$ written through $\to$, $\bot$), or got from two earlier ones B and $B \to A$. Write $\vdash A$. Checking a proof is a recursion; finding one is a search.

D3 A *world*: an arrow from the letters to $\{0, 1\}$. It gives every statement a value, by recursion ($\bot \mapsto 0$, and the usual tables).

Blur

The floor: statements of D1. O: $\neg$, $\wedge$, $\vee$, $\to$.

B1 (From below, L5: forget the proof.) Gluings: $A \sim B$ when $\vdash A \leftrightarrow B$. Respects: $\neg$, $\wedge$, $\vee$, $\to$.

B2 (From above, L6: forget all but the answers.) Questions: for each world w, the value of a statement in w. Respects: $\neg$, $\wedge$, $\vee$, $\to$.

Propositions

P1 (Survives.) $\neg$, $\wedge$, $\vee$, $\to$ respect B1 (from $\vdash A \leftrightarrow A'$ and $\vdash B \leftrightarrow B'$, prove $\vdash (A \wedge B) \leftrightarrow (A' \wedge B')$ and the others) and B2 (the tables are taken world by world).

P2 (Dies: length.) p_1 and $\neg\neg p_1$ are glued; their lengths are 1 and 3. "Which proof" is the first thing forgotten, and "how short a proof can be" goes with it (R1).

P3 (Sound: Z4.) If $A \sim B$ then A, B have the same value in every world, in every context. (Each axiom is 1 in every world, and "from B and $B \to A$, get A" keeps "1 in every world"; then P1.)

P4 (A construction.) If A is 1 in every world, then $\vdash A$. (For each world w, prove "the row of $w \to A$" by recursion on A; then remove the letters one at a time with $\vdash (p \to A) \to ((\neg p \to A) \to A)$.)

P5 (Appears.) $\sim\ =$ B2. (Z4 by P3. Z5: if $A \not\sim B$, then $A \leftrightarrow B$ is not provable, so by P4 some world w gives it 0; the separating context is the empty one, and the question is "the value in w".)

P6 (Normal forms.) Each bag holds exactly one "or" of full rows, one row for each world giving 1 (rows and letters in a fixed order; the empty "or" is $\bot$).

Notation

N1 Write $[A]$ for the bag of A, and $\mathcal{B}_n$ for the bags (M5), with $\neg$, $\wedge$, $\vee$, $\to$ carried over by P1 (Z6). Let $\top := [\bot \to \bot]$.

N2 "A is provable" is now "$[A] = \top$".

Arrow

P7 $\mathcal{B}_n \to \{0,1\}^{\text{worlds}}$, $[A] \mapsto$ (the value of A in each world). Each of $\neg$, $\wedge$, $\vee$, $\to$: lift (D1), survive (P1), descend (Z6), agree (the tables, world by world). One-to-one by P5 and onto by P6: the arrow is *inside* and a *picture* at once (W2).

Theorems

T1 (Count.) There are exactly 2^{2^n} bags. (P6, P7.)

T2 (Provable is true.) $\vdash A \iff A$ is 1 in all 2^n worlds. So provability is decided by a finite check, and the check is not a proof (E3).

T3 (Laws.) $\mathcal{B}_n$ with $\wedge$, $\vee$, $\neg$ obeys every equation that $\{0,1\}$ obeys. (Equations: free by Z7 through P7.)

Remarks

R1 (How much proof.) Two sizes die in B1: *length*, the shortest proof, and *strength*, the first rung of a ladder of axiom lists that proves A. For the pigeonhole statement with $n+1$ birds the shortest proofs in weak systems grow faster than any power of n (reported, classical; no line depends on it). Counting bags makes it a few lines (Building 1, Chapter II.6).

R2 (Reported: with "for all". No line of this building depends on it.) For statements about a floor with "for all" and "there is", and a list of axioms Γ, B1 (each proves the other from Γ) and B2 (same value in every floor obeying Γ) agree. The proof builds a floor out of the words themselves.

R3 (Reported: Appears fails for one world. No line of this building depends on it.) Take the axioms of $\mathbb{N}$ with $+$, $\times$, and replace B2 by the single world $\mathbb{N}$. Then B1 is strictly finer: some G is true in $\mathbb{N}$ and not glued to $\top$. Z4 holds, Z5 fails, with G as witness. The proof codes words as numbers, so that statements about $\mathbb{N}$ speak about statements: a loop of the kind this building is named for. Chapter 2 builds that loop as a floor of its own.

R4 (A twist.) A statement "for every n, a check on n says yes" is refuted by one finite witness if it is false. So if it is shown to be neither provable nor refutable, it is true in $\mathbb{N}$.

R5 (Old names.) D1: propositional formulas; D2: a Hilbert system; D3: valuations; $\mathcal{B}_n$: the Lindenbaum algebra, the free Boolean algebra on n generators; P3: soundness; P4, P5: completeness (Post; Kalmár's proof); R2: Gödel's completeness theorem; R3: Gödel's first incompleteness theorem.

R6 (Machine check, `checks/b2-ch1-2-check.py`.) The axioms on many instances are 1 in every world; $2^{2^2} = 16$ bags for two letters; P6 normal forms.

Exercises

E1 Write the normal form of P6 for $p_1 \to p_2$ and for $\neg(p_1 \leftrightarrow p_2)$.

E2 Show $\vdash p_1 \to p_1$ from the axioms of D2 in five lines.

E3 Why does T2 not give a fast way to find proofs?

Chapter 2. Equally provable

box statements → how provable they are · from: Chapter 1; Building 0, Chapters 0, 1 · build, blur

Add one sign, $\Box$, to statements without letters, and the rules a word "provable" would have to obey. Glue two statements when the rules prove "this one is provable exactly when that one is". The bags are a ladder of shape $\omega + 1$, and the rung of a statement is the first depth at which it fails: a valuation, like v_p.

Given

G1 (Chapter 1.) Statements, proofs from the three axioms and "from B and $B \to A$, get A"; worlds; P3 (sound).

G2 (Building 0, Chapter 1.) $\mathbb{N}$ with $<$, and recursion on $\mathbb{N}$.

Definitions

D1 (Build, M3.) *Box statements*: words in $\bot$, $\to$, $\Box$ (no letters), with $\neg$, $\wedge$, $\vee$, $\leftrightarrow$ written through $\to$, $\bot$ as in Chapter 1. Write $\Box^0\bot := \bot$, $\Box^{k+1}\bot := \Box\Box^k\bot$, and $\top := \bot \to \bot$.

D2 (Build.) $\vdash A$: A is the end of a list in which each statement is an instance of Chapter 1's three axioms, or of

$$\text{(K)}\ \ \Box(A \to B) \to (\Box A \to \Box B), \qquad \text{(L)}\ \ \Box(\Box A \to A) \to \Box A,$$

or follows from earlier ones by "from B and $B \to A$, get A" or "from A, get $\Box A$".

D3 (Recursion on $\mathbb{N}$, G2.) The value of A at depth n: $\bot$ is 0; $\to$ by its table; $\Box A$ is 1 at depth n when A is 1 at every depth $m < n$.

Blur

The floor: box statements. O: $\wedge$, $\Box$.

B1 (From below, L5.) Gluings: $A \approx B$ when $\vdash \Box A \leftrightarrow \Box B$. Respects: $\wedge$, $\Box$.

B2 (From above, L6.) Question: the first depth at which A is 0, or ∞ if there is none. Respects: $\wedge$, $\Box$.

Propositions

P1 (Sound.) If $\vdash A$ then A is 1 at every depth. (At a fixed depth, treat each $\Box X$ as a letter: Chapter 1's axioms are 1 there. K: if $A \to B$ and A are 1 below n, so is B. L: suppose $\Box A \to A$ is 1 at every $m < n$; if A is 1 at every $k < m$ then $\Box A$ is 1 at m, so A is; by recursion on m, A is 1 below n. Both rules keep "1 at every depth".)

P2 (Read off the question.) Write $\rho(A)$ for the answer of B2. Then $\Box A$ is 1 at depth n exactly when $n \le \rho(A)$. Hence $\rho(\Box A) = \rho(A) + 1$, $\rho(A \wedge B) = \min(\rho(A), \rho(B))$, and $\rho(\Box^k\bot) = k$.

P3 (Survives.) $\wedge$ and $\Box$ respect B1: $\vdash \Box(A \wedge B) \leftrightarrow \Box A \wedge \Box B$, and from $\vdash X \leftrightarrow Y$ get $\vdash \Box X \leftrightarrow \Box Y$ (rule and K). They respect B2 by P2.

P4 (Steps go one way.) $\vdash \Box^i \bot \to \Box^j \bot$ when $i \leq j$. ($\vdash \bot \to \Box^{j-i} \bot$; then the rule and K, i times.)

P5 (Four shapes.) By P4, every "or" of steps $\Box^i \bot$ and negated steps $\neg\Box^j \bot$ is provably $\top$, or one of $\Box^b \bot$; $\neg\Box^a \bot$; $\Box^a \bot \to \Box^b \bot$ with $b < a$; $\bot$. Their boxes:

$$\vdash \Box\Box^b \bot \leftrightarrow \Box^{b+1} \bot, \qquad \vdash \Box\neg\Box^a \bot \leftrightarrow \Box \bot \quad (a \geq 1),$$
$$\vdash \Box(\Box^a \bot \to \Box^b \bot) \leftrightarrow \Box^{b+1} \bot \quad (b < a).$$

(The second: L with $A = \bot$, since $\neg\Box^a \bot \to \neg\Box \bot$ by P4. The third, one way: $\Box^a \bot \to \Box^b \bot$ gives $\Box^{b+1} \bot \to \Box^b \bot$ by P4, box it, and L with $A = \Box^b \bot$. The other way: $\Box^b \bot$ gives the implication, box it.)

P6 (Normal form; a construction.) For every A: $\vdash \Box A \leftrightarrow \Box^c \bot$ for one $c \geq 1$, or $\vdash \Box A$. (By recursion on A, every box statement is provably a Boolean word in steps; put it as an "and" of "or"s, box it with P3, and use P5; an "and" of steps is its lowest step, P4.)

P7 (Appears.) $\approx =$ B2. (Z4: if $A \approx B$, then by P1 $\Box A$ and $\Box B$ agree at every depth, so $\rho(A) = \rho(B)$ by P2, in every context by P3. Z5: by P6 and P1, $\vdash \Box A \leftrightarrow \Box^c \bot$ forces $c = \rho(A) + 1$, so if $A \not\approx B$ their rungs differ, say $\rho(A) = k < \rho(B)$; the separating context is $X \mapsto \Box X$ and the question is "the value at depth $k+1$": 0 for $\Box A$, 1 for $\Box B$.)

P8 (Dies: the value at depth 1.) $\bot \approx \neg\Box \bot$ (both on rung 0, P2), but at depth 1, $\bot$ is 0 and $\neg\Box \bot$ is 1.

P9 (Dies: $\vee$ and $\neg$.) $\bot \approx \neg\Box \bot$, but $\bot \vee \Box \bot$ is on rung 1 while $\neg\Box \bot \vee \Box \bot$ is on rung ∞.

Notation

N1 The bags are named by their rungs: $0, 1, 2, \ldots, \infty$ (P7). Call $\rho(A)$ the *rung* of A.

N2 $\Box$ and $\wedge$ carried over (P3, Z6): on rungs they are $k \mapsto k+1$ and min (P2).

Arrow

P10 Bags $\to \mathbb{N} \cup \{\infty\}$, $[A] \mapsto \rho(A)$. $\Box$: lift (D1), survive (P3), descend (Z6), agree (+1). $\wedge$: the same, agreeing with min. One-to-one (P7) and onto (P2): inside and a picture.

Theorems

T1 (A ladder of shape $\omega + 1$.) The rungs are $0 < 1 < 2 < \cdots < \infty$, every one occurs, and $\Box^k \bot$ is a normal form of rung k, $\top$ of rung ∞.

T2 (Löb, inside.) $A \approx (\Box A \to A)$ for every A. ($\vdash \Box A \to \Box(\Box A \to A)$ by K; L the other way.)

T3 (Two rungs never glued, straight from L.) $\vdash \Box^{k+1} \bot \to \Box^k \bot$ never holds: boxing it and L would give $\vdash \Box^{k+1} \bot$, against P1.

T4 (Placement.) $\bot$ and $\neg\Box\bot$ on rung 0; $\Box\bot$ on rung 1; $\Box\bot \to \Box\Box\bot$ on rung ∞.

Remarks

R1 (The twin.) P2 is the law of v_p (Building 0, Chapter 18) and of ord_a (Building 0, Chapter 34) with $\times$ read as $\wedge$: the rung is the first depth at which a statement stops being glued to $\top$, as $v_p(a)$ is the first rung at which a stops being glued to 0. Here the ladder ends at $\omega + 1$ instead of running on.

R2 (Reported: the arrow into arithmetic. No line of this chapter depends on it.) Read $\Box A$ as "A is provable from the axioms of $\mathbb{N}$". That this arrow respects D2 (the conditions on $\Box$ and Löb's theorem), and that it glues no more than D2 does on box statements (the depth reading), are classical. Under it, P8 says: "the axioms are consistent" ($\neg\Box\bot$) is true and glued to $\bot$, so it is not provable. Truth dies in the blur; that is the whole content of the second incompleteness theorem seen from here.

R3 (What is known.) P6, P7 are the letterless normal form of the logic of $\Box$ (Boolos). Stronger boxes stretch the ladder to the ordinal ε_0 (Beklemishev; reported, no line depends on it). The blur B1 with letters is not worked out: open.

R4 (Old names.) D2: the logic GL; K: distribution; L: Löb's axiom; D3: the frame of the natural numbers read backwards; R2: the Hilbert–Bernays–Löb conditions, Löb's theorem, Solovay's completeness, Gödel's second incompleteness theorem.

R5 (Where existence came from.) Every bag exists by M5. That every rung occurs comes from a Build ($\Box^k\bot$), and the normal form from recursion on words (P6).

R6 (Machine check, `checks/b2-ch1-2-check.py`.) On box statements up to size 3 read at depths below 40: P2, P5's three boxes, P7 (bags are rungs), T2, T4, P9.

Exercises

E1 Find the rung of $\Box\bot \to \bot$ and of $\Box\Box\bot \wedge \neg\Box\bot$.

E2 Why is $\vee$ not max on rungs? Find $\rho(A \vee B)$ for $A = \bot$, $B = \Box\bot$.

E3 Use P5 to find the normal form of $\Box(\neg\Box\Box\bot \vee \Box\bot)$.